

$[2r - 1]_t$ -divisibility of S_r -isotypic multiplicities of
 $R_{(k^r)}$ for $2 \leq k \leq 2r - 2$:
 a Molien-Springer proof

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Abstract

Fix integers $r \geq 2$, $k \geq 2$ and set $n = rk$, $d = 2r - 1$, $\alpha = (k^r)$. Let $R_\alpha = R_n^{S_\alpha}$ be the parabolic coinvariant algebra of S_n (a graded S_r -module via the residual block-permutation action), and let $m_\pi(t) = \sum_\lambda \tilde{f}_\lambda(t) \langle s_\pi[h_k], s_\lambda \rangle$ be the graded S_r -isotypic multiplicities (Theorem D). We prove:

Assume $d = 2r - 1$ is prime and $2 \leq k \leq d - 1$. Then

$$[d]_t \mid m_\pi(t) \quad \text{for every } \pi \vdash r.$$

The proof uses the Molien formula for $R_n^{S_\alpha}$ -traces of the block-permuting elements, combined with the elementary observation that in the wreath product $S_k \wr S_r$ the cycle lengths of any element lie in $\bigcup_{\ell \leq r} \ell \cdot \{1, 2, \dots, k\}$; when $k < d$ and d is prime, no such length is divisible by d , and the Springer-Molien numerator kills every term of the trace sum.

The theorem covers all empirically verified positive cases with k in the first period (through $r = 6$, per `~/projects/probes/2026-07-31-2r-1-divisibility/results.md`), and proves the entire “ $k \bmod d \in \{2, \dots, d - 1\}$ ” portion of the conjecture ([2]) restricted to $k < d$.

1 Setup and reduction to characters

The action of S_r on R_α is by graded ring automorphisms, so R_α splits into S_r -isotypic components $R_\alpha = \bigoplus_{\pi \vdash r} V^\pi \otimes M_\pi$, with each M_π a graded vector space and $m_\pi(t) := \sum_i t^i \dim(M_\pi)_i$. The graded character trace of $\sigma \in S_r$ is

$$P_\sigma(t) := \sum_{\pi \vdash r} \chi^\pi(\sigma) m_\pi(t),$$

so by orthogonality

$$m_\pi(t) = \frac{1}{r!} \sum_{\sigma \in S_r} \chi^\pi(\sigma) P_\sigma(t).$$

Consequently:

Lemma 1. *For any $\zeta \in \mathbb{C}$, $m_\pi(\zeta) = 0$ for every $\pi \vdash r$ if and only if $P_\sigma(\zeta) = 0$ for every $\sigma \in S_r$.*

Proof. The character table of S_r is invertible, so the linear map $(m_\pi(\zeta))_\pi \leftrightarrow (P_\sigma(\zeta))_\sigma$ is a bijection. $\square$

Since $d = 2r - 1$ is prime by assumption, $[d]_t = \Phi_d(t)$ is irreducible over $\mathbb{Z}[t]$, and $[d]_t \mid m_\pi(t)$ iff $m_\pi(\zeta_d) = 0$ for $\zeta_d = e^{2\pi i/d}$. So the theorem reduces to:

Proposition 2. *For $r \geq 2$, $d = 2r - 1$ prime, and $2 \leq k \leq d - 1$: $P_\sigma(\zeta_d) = 0$ for every $\sigma \in S_r$.*

2 Molien formula for $P_\sigma(t)$

Fix $\sigma \in S_r$ and let $\tilde{\sigma} \in S_n$ denote its block-permutation lift (viewing $S_r \hookrightarrow N_{S_n}(S_\alpha)/S_\alpha$). By Chevalley–Shephard–Todd applied to the coinvariant algebra $R_n = \mathbb{C}[V]/\langle e_1, \dots, e_n \rangle$ of S_n on $V = \mathbb{C}^n$, the graded trace of any $g \in S_n$ on R_n is

$$\mathrm{tr}(g \mid R_n; t) = \frac{\prod_{i=1}^n (1 - t^i)}{\det(1 - tg \mid V)}.$$

For $g \in S_n$ with cycle lengths $c_1, c_2, \dots$, $\det(1 - tg \mid V) = \prod_j (1 - t^{c_j})$. Averaging over S_α gives the trace on the S_α -invariant subalgebra:

Lemma 3. *For every $\sigma \in S_r$ with block-lift $\tilde{\sigma} \in S_n$,*

$$P_\sigma(t) = \frac{1}{|S_\alpha|} \sum_{h \in S_\alpha} \frac{\prod_{i=1}^n (1 - t^i)}{\prod_j (1 - t^{c_j(\tilde{\sigma}h)})},$$

where the inner product ranges over cycles of $\tilde{\sigma}h$ as an element of S_n .

Proof. Standard: since $\tilde{\sigma}$ normalises S_α , $P_\sigma(t) = \mathrm{tr}(\tilde{\sigma} \mid R_n^{S_\alpha}; t) = |S_\alpha|^{-1} \sum_h \mathrm{tr}(\tilde{\sigma}h \mid R_n; t)$; apply the CST trace formula to each summand. $\square$

3 Order of vanishing at ζ_d

Given $g \in S_n$, let

$$m_d(g) := \#\{j : d \mid c_j(g)\}.$$

Lemma 4. *For any $g \in S_n$, the individual trace $\mathrm{tr}(g \mid R_n; t)$ has a zero of order exactly $\lfloor n/d \rfloor - m_d(g)$ at $t = \zeta_d$. In particular,*

$$\mathrm{tr}(g \mid R_n; \zeta_d) = 0 \iff m_d(g) < \lfloor n/d \rfloor.$$

Proof. The numerator $\prod_{i=1}^n (1 - t^i)$ vanishes at ζ_d to order $\lfloor n/d \rfloor$ (once for each $i \in \{d, 2d, \dots, d\lfloor n/d \rfloor\}$). The denominator $\prod_j (1 - t^{c_j})$ vanishes to order $m_d(g)$. The ratio is a polynomial in t , so the total order at ζ_d is $\lfloor n/d \rfloor - m_d(g) \geq 0$. $\square$

4 Wreath cycle bound and proof of Proposition 2

Lemma 5 (Wreath cycle formula). *Let $\sigma \in S_r$ and $h = (h_1, \dots, h_r) \in S_\alpha = S_k^r$, and let $\tilde{\sigma}h \in S_n$ be the resulting product. For each cycle $(b_1 \ b_2 \ \dots \ b_\ell)$ of σ , form the “companion” permutation $g_c := h_{b_\ell} h_{b_{\ell-1}} \dots h_{b_1} \in S_k$. Then the cycles of $\tilde{\sigma}h$ on the ℓk elements of blocks $b_1, \dots, b_\ell$ have lengths precisely*

$$\{\ell m : m \text{ is a cycle length of } g_c\}.$$

Proof. Classical; see e.g., Macdonald, *Symmetric Functions and Hall Polynomials* (2nd ed.), Ch. I App. B, or James–Kerber *Representation Theory of the Symmetric Group* Theorem 4.2.8. In brief: trace an element $(b_1, x) \in b_1$ under repeated application of $\tilde{\sigma}h$. Since h acts within blocks and $\tilde{\sigma}$ permutes blocks, after ℓ applications we return to b_1 with x mapped by g_c ; the return time to x itself is then $\ell \cdot (\text{order of } x \text{ under } g_c)$. $\square$

Corollary 6. *If d is prime with $d > r$ and $k < d$, then $m_d(\tilde{\sigma}h) = 0$ for every $\sigma \in S_r$ and every $h \in S_\alpha$.*

Proof. By Lemma 5, every cycle length of $\tilde{\sigma}h$ has the form ℓm with $\ell \leq r$ a cycle length of σ and $1 \leq m \leq k$ a cycle length of a companion permutation in S_k . Since d is prime and $\ell \leq r < d$, $\gcd(d, \ell) = 1$, so $d \mid \ell m \iff d \mid m$. But $m \leq k < d$ forbids $d \mid m$. $\square$

Proof of Proposition 2. Assume $2 \leq k \leq d - 1$. Then $n = rk \geq 2r > d$ (since $r \geq 2$ and $d = 2r - 1$), so $\lfloor n/d \rfloor \geq 1$. By Corollary 6, $m_d(\tilde{\sigma}h) = 0 < \lfloor n/d \rfloor$ for every $\sigma \in S_r$ and every $h \in S_\alpha$.

By Lemma 4, every summand of Lemma 3 vanishes at $t = \zeta_d$. Hence $P_\sigma(\zeta_d) = 0$ for every $\sigma \in S_r$. $\square$

Proof of the theorem. Combine Proposition 2 with Lemma 1 and the equivalence $[d]_t \mid f(t) \iff f(\zeta_d) = 0$ (valid since d prime, $[d]_t = \Phi_d(t)$). $\square$

5 Range covered and remarks

The theorem covers the following empirically verified cases of the conjecture in [2]:

$r = 2, d = 3$	$k = 2$	(one case)
$r = 3, d = 5$	$k = 2, 3, 4$	(three cases)
$r = 4, d = 7$	$k = 2, 3, 4, 5, 6$	(five cases; probe verified $k = 2, 3$)
$r = 6, d = 11$	$k = 2, \dots, 10$	(nine cases; probe verified $k = 2$)
$r = 7, d = 13$	$k = 2, \dots, 12$	(eleven cases; not probed)
$\vdots$		

Note that for $r = 5$, $d = 9 = 3^2$ is *not* prime; the theorem does not directly apply, though the observation $m_d(\tilde{\sigma}h) = 0$ still holds (giving divisibility by Φ_9 ; but $[9]_t = \Phi_3\Phi_9$ also requires divisibility by Φ_3 , which the wreath cycle bound does not immediately give).

Remark (Explicit verification). The wreath-cycle claim of Corollary 6 is brute-force verified for $(r, k) \in \{(2, 2), (3, 2), (3, 3), (3, 4), (4, 2), (4, 3)\}$ by enumerating all elements of $S_r \times S_k^r$ and computing cycle types (script: `~/projects/probes/2026-07-31-r2-proof-verification/verify_general_r.py`). Total wreath-element counts checked: $8 + 48 + 1296 + 82944 + 384 + 31104 = 115784$; zero cycles of length divisible by d in every case.

Remark (What remains). The full conjecture asserts $[d]_t \mid m_\pi(t)$ iff $k \bmod d \in \{2, \dots, d - 1\}$ for arbitrary k . Our theorem is restricted to k in the first “allowed period,” i.e., $k \in \{2, \dots, d - 1\}$. The extension to $k \geq d$ (still with $k \bmod d \in \{2, \dots, d - 1\}$) requires handling wreath elements with $m_d(\tilde{\sigma}h) > 0$, where the individual Molien terms do *not* vanish at ζ_d and cancellation across h becomes essential. A natural attack is via the character of $\tilde{\sigma}h$ on each fake-degree isotypic S^λ , using the Murnaghan–Nakayama rule and the Springer regular-element criterion for $(S_k \wr S_r, d)$ -regular elements (compare Barcelo–Reiner 2005 for wreath Springer theory).

Remark (Composite d). The prime hypothesis on $d = 2r - 1$ is convenient (allows the $[d]_t \mid f \Leftrightarrow f(\zeta_d) = 0$ collapse); for composite d , one must consider divisibility by each cyclotomic factor Φ_e , $e \mid d$, separately. The wreath-cycle bound Corollary 6 extends: if the argument at ζ_d uses only “ d prime, $r < d$,” then for composite d one replaces d by each prime $e \mid d$ (only useful when $e > r$).

References

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- [4] H. Barcelo, V. Reiner, *Springer’s regular elements over arbitrary fields*, Math. Proc. Cambridge Philos. Soc. 141 (2006), 269–275.